

```
from google.colab import drive
drive.mount('/content/drive')

import os
PROJECT_DIR = "/content/drive/MyDrive/PRISM AI"
os.makedirs(f"{PROJECT_DIR}/adapters", exist_ok=True)
```

Mounted at /content/drive

```
DATASET_DIR = f"{PROJECT_DIR}/datasets" # where you uploaded train.jsonl / val.jsonl
ADAPTER_OUTPUT_PATH = f"{PROJECT_DIR}/adapters/prism_lora" # ONE adapter, not one per format
```

```
!pip install --quiet unsloth
!pip install --quiet --no-deps trl peft accelerate bitsandbytes
!pip install --upgrade fsspec==2025.12.0
!pip install --upgrade gcsfs==2025.9.0
!pip check
```

```
Installing collected packages: fsspec
  Attempting uninstall: fsspec
    Found existing installation: fsspec 2025.9.0
    Uninstalling fsspec-2025.9.0:
      Successfully uninstalled fsspec-2025.9.0
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour
datasets 4.3.0 requires fsspec[http]<=2025.9.0,>=2023.1.0, but you have fsspec 2025.12.0 which is incompatible.
Successfully installed fsspec-2025.12.0
Collecting gcsfs==2025.9.0
  Downloading gcsfs-2025.9.0-py2.py3-none-any.whl.metadata (2.1 kB)
Requirement already satisfied: aiohttp!=4.0.0a0,!4.0.0a1 in /usr/local/lib/python3.13/dist-packages (from gcsfs==2025.9.0)
Requirement already satisfied: decorator>4.1.2 in /usr/local/lib/python3.13/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1) (4.4.2)
Collecting fsspec==2025.9.0 (from gcsfs==2025.9.0)
  Using cached fsspec-2025.9.0-py3-none-any.whl.metadata (10 kB)
Requirement already satisfied: google-auth>=1.2 in /usr/local/lib/python3.13/dist-packages (from gcsfs==2025.9.0) (2.49.0)
Requirement already satisfied: google-auth-oauthlib in /usr/local/lib/python3.13/dist-packages (from gcsfs==2025.9.0) (1.4)
Requirement already satisfied: google-cloud-storage in /usr/local/lib/python3.13/dist-packages (from gcsfs==2025.9.0) (3.1)
Requirement already satisfied: requests in /usr/local/lib/python3.13/dist-packages (from gcsfs==2025.9.0) (2.32.4)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.13/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1)
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.13/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1)
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.13/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1)
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.13/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1)
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.13/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1)
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.13/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1)
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.13/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1)
Requirement already satisfied: pyasn1-modules>=0.2.1 in /usr/local/lib/python3.13/dist-packages (from google-auth>=1.2->gc
Requirement already satisfied: cryptography>=38.0.3 in /usr/local/lib/python3.13/dist-packages (from google-auth>=1.2->gc
Requirement already satisfied: rsa<5,>=3.1.4 in /usr/local/lib/python3.13/dist-packages (from google-auth>=1.2->gcsfs==202
Requirement already satisfied: requests-oauthlib>=0.7.0 in /usr/local/lib/python3.13/dist-packages (from google-auth-oauth
Requirement already satisfied: google-api-core<3.0.0,>=2.27.0 in /usr/local/lib/python3.13/dist-packages (from google-clou
Requirement already satisfied: google-cloud-core<3.0.0,>=2.4.2 in /usr/local/lib/python3.13/dist-packages (from google-clo
Requirement already satisfied: google-resumable-media<3.0.0,>=2.7.2 in /usr/local/lib/python3.13/dist-packages (from googl
Requirement already satisfied: google-crc32c<2.0.0,>=1.6.0 in /usr/local/lib/python3.13/dist-packages (from google-cloud-s
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.13/dist-packages (from requests->gcsfs==
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.13/dist-packages (from requests->gcsfs==2025.9.0) (3)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.13/dist-packages (from requests->gcsfs==2025.9
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.13/dist-packages (from requests->gcsfs==2025.9
Requirement already satisfied: cffi>=2.0.0 in /usr/local/lib/python3.13/dist-packages (from cryptography>=38.0.3->google-a
Requirement already satisfied: googleapis-common-protos<2.0.0,>=1.63.2 in /usr/local/lib/python3.13/dist-packages (from go
Requirement already satisfied: protobuf<8.0.0,>=4.25.8 in /usr/local/lib/python3.13/dist-packages (from google-api-core<3
Requirement already satisfied: proto-plus<2.0.0,>=1.22.3 in /usr/local/lib/python3.13/dist-packages (from google-api-core<
Requirement already satisfied: pyasn1<0.7.0,>=0.6.1 in /usr/local/lib/python3.13/dist-packages (from pyasn1-modules>=0.2.1
Requirement already satisfied: oauthlib>=3.0.0 in /usr/local/lib/python3.13/dist-packages (from requests-oauthlib>=0.7.0->
Requirement already satisfied: pycparser in /usr/local/lib/python3.13/dist-packages (from cffi>=2.0.0->cryptography>=38.0.
Downloading gcsfs-2025.9.0-py2.py3-none-any.whl (36 kB)
Using cached fsspec-2025.9.0-py3-none-any.whl (199 kB)
Installing collected packages: fsspec, gcsfs
  Attempting uninstall: fsspec
    Found existing installation: fsspec 2025.12.0
    Uninstalling fsspec-2025.12.0:
      Successfully uninstalled fsspec-2025.12.0
  Attempting uninstall: gcsfs
    Found existing installation: gcsfs 2025.12.0
    Uninstalling gcsfs-2025.12.0:
      Successfully uninstalled gcsfs-2025.12.0
Successfully installed fsspec-2025.9.0 gcsfs-2025.9.0
ipython 7.34.0 requires jedi, which is not installed.
```

```
from unsloth import FastLanguageModel

MAX_SEQ_LENGTH = 2048

model, tokenizer = FastLanguageModel.from_pretrained(
```

```

model_name="unsloth/Qwen2.5-7B-Instruct-bnb-4bit",
max_seq_length=MAX_SEQ_LENGTH,
dtype=None,
load_in_4bit=True,
)

```

🐍 Unsloth: Will patch your computer to enable 2x faster free finetuning.
 🐍 Unsloth Zoo will now patch everything to make training faster!
 ==((===))== Unsloth 2026.9.8: Fast Qwen2 patching. Transformers: 5.5.0.
 \ \ / / Tesla T4. Num GPUs = 1. Max memory: 14.563 GB. Platform: Linux.
 0^0/ _/ \ Torch: 2.11.0+cu128. CUDA: 7.5. CUDA Toolkit: 12.8. Triton: 3.6.0
 \ _/ / Bfloat16 = FALSE. FA [Xformers = 0.0.35. FA2 = False]
 "_____" Free license: <http://github.com/unslothai/unsloth>
 Unsloth: Fast downloading is enabled - ignore downloading bars which are red colored!
 Loading weights: 100% 339/339 [00:19<00:00, 24.69it/s]

```

model = FastLanguageModel.get_peft_model(
    model,
    r=16,
    target_modules=["q_proj", "k_proj", "v_proj", "o_proj",
                   "gate_proj", "up_proj", "down_proj"],
    lora_alpha=16,
    lora_dropout=0,
    bias="none",
    use_gradient_checkpointing="unsloth",
    random_state=42,
)

```

Unsloth 2026.9.8 patched 28 layers with 28 QKV layers, 28 O layers and 28 MLP layers.

Double-click (or enter) to edit

```

from datasets import load_dataset

FORMAT_INSTRUCTIONS = {
    "advisory": "You are a security advisory writer. Convert the given source content into a structured advisory with sections.",
    "linkedin": "You are a social media specialist. Convert the given source content into a professional LinkedIn post.",
    "twitter": "You write platform-optimized social content. Convert the given source content into a short Twitter/X post or thread (2-4 tweets).",
    "video": "You write video scripts. Convert the given source content into a video package: script, storyboard beats, scene descriptions.",
    "infographic": "You design infographics. Convert the given source content into infographic content: key messaging points, visual structure.",
    "executive_summary": "You write executive briefings. Convert the given source content into a concise executive summary.",
    "presentation": "You create presentations. Convert the given source content into presentation slide content (titles + bullet points).",
}

def formatting_func(example):
    instruction = FORMAT_INSTRUCTIONS[example["format"]] # <-- looked up per example
    messages = [
        {"role": "system", "content": instruction},
        {"role": "user", "content": example["input"]},
        {"role": "assistant", "content": example["output"]},
    ]
    text = tokenizer.apply_chat_template(messages, tokenize=False, add_generation_prompt=False)
    return {"text": text}

train_dataset = load_dataset("json", data_files=f"{DATASET_DIR}/train.jsonl", split="train")
val_dataset = load_dataset("json", data_files=f"{DATASET_DIR}/val.jsonl", split="train")

train_dataset = train_dataset.map(formatting_func)
val_dataset = val_dataset.map(formatting_func)

print(f"Train: {len(train_dataset)} examples across all formats")
print(f"Val: {len(val_dataset)} examples")
print("--- Example ---")
print(train_dataset[0]["format"], "->", train_dataset[0]["text"][:400])

```

Generating train split: 1260/0 [00:02<00:00, 607.88 examples/s]

Generating train split: 139/0 [00:00<00:00, 175.48 examples/s]

Map: 100% 1260/1260 [00:00<00:00, 7066.33 examples/s]

Map: 100% 139/139 [00:00<00:00, 1960.63 examples/s]

Train: 1260 examples across all formats

Val: 139 examples

--- Example ---

twitter -> <|im_start|>system

You write platform-optimized social content. Convert the given source content into a short Twitter/X post or thread (2-4 tweets).

<|im_start|>user

Global logistics firm TransLogistics reports a significant bottleneck at the Port of Rotterdam, affecting approximately 4,500 containers per day.

B7. Train

```
#train
from trl import SFTTrainer, SFTConfig
```

```
training_args = SFTConfig(
    output_dir="/content/prism_checkpoints",
    per_device_train_batch_size=2,
    gradient_accumulation_steps=4,
    num_train_epochs=3,
    learning_rate=2e-4,
    logging_steps=10,
    eval_strategy="epoch",
    save_strategy="epoch",
    optim="adamw_8bit",
    warmup_steps=10,
    max_seq_length=MAX_SEQ_LENGTH,
    dataset_text_field="text",
    report_to="none",
)
```

```
trainer = SFTTrainer(
    model=model,
    tokenizer=tokenizer,
    train_dataset=train_dataset,
    eval_dataset=val_dataset,
    args=training_args,
)
```

```
trainer.train()
```

```
Unsloth: Tokenizing ["text"] (num_proc=2): 100%
```

```
1260/1260 [00:19<00:00, 66.78 examples/s]
```

```
Unsloth: Tokenizing ["text"] (num_proc=2): 100%
```

```
139/139 [00:06<00:00, 22.69 examples/s]
```

The tokenizer has new PAD/BOS/EOS tokens that differ from the model config and generation config. The model config and generation config will be updated to match the tokenizer config. Unsloth: Padding-free auto-enabled, enabling faster training.

```
==((====))== Unsloth - 2x faster free finetuning | Num GPUs used = 1
\\  /      Num examples = 1,260 | Num Epochs = 3 | Total steps = 474
0'0/  \    Batch size per device = 2 | Gradient accumulation steps = 4
 \  _/      Data Parallel GPUs = 1 | Total batch size (2 x 4 x 1) = 8
"-__-"      Trainable parameters = 40,370,176 of 7,655,986,688 (0.53% trained)
```

```
`use_return_dict` is deprecated! Use `return_dict` instead!
```

```
Unsloth: Will smartly offload gradients to save VRAM!
```

```
/usr/local/lib/python3.13/dist-packages/unsloth/import_fixes.py:3239: UserWarning: 'has_cuda' is deprecated, please use 'torch.cuda.is_available()'
return original(name)
```

```
/usr/local/lib/python3.13/dist-packages/unsloth/import_fixes.py:3239: UserWarning: 'has_cudnn' is deprecated, please use 'torch.backends.cuda.is_available()'
return original(name)
```

```
/usr/local/lib/python3.13/dist-packages/unsloth/import_fixes.py:3239: UserWarning: 'has_mps' is deprecated, please use 'torch.backends.mps.is_available()'
return original(name)
```

```
/usr/local/lib/python3.13/dist-packages/unsloth/import_fixes.py:3239: UserWarning: 'has_mkl_dnn' is deprecated, please use 'torch.backends.mkl.is_available()'
return original(name)
```

```
Unsloth: Double buffering enabled (parallel H2D + compute) for backward pass.
```

```
[474/474 1:48:34, Epoch 3/3]
```

Epoch	Training Loss	Validation Loss
-------	---------------	-----------------

1	0.878291	0.875709
---	----------	----------

2	0.736453	0.842530
---	----------	----------

3	0.622680	0.861173
---	----------	----------

```
Unsloth: Restored added_tokens_decoder metadata in /content/prism_checkpoints/checkpoint-158/tokenizer_config.json.
```

```
Unsloth: Restored added_tokens_decoder metadata in /content/prism_checkpoints/checkpoint-316/tokenizer_config.json.
```

```
Unsloth: Restored added_tokens_decoder metadata in /content/prism_checkpoints/checkpoint-474/tokenizer_config.json.
```

```
TrainOutput(global_step=474, training_loss=0.7948463013403526, metrics={'train_runtime': 6540.0312,
'train_samples_per_second': 0.578, 'train_steps_per_second': 0.072, 'total_flos': 8.994917932406784e+16, 'train_loss':
0.7948463013403526, 'epoch': 3.0})
```

B8. Save the adapter — one file set, not seven

```
model.save_pretrained(ADAPTER_OUTPUT_PATH)
tokenizer.save_pretrained(ADAPTER_OUTPUT_PATH)
print("Saved to", ADAPTER_OUTPUT_PATH)
```

```
Unsloth: Restored added_tokens_decoder metadata in /content/drive/MyDrive/PRISM AI/adapters/prism_lora/tokenizer_config.json.
Saved to /content/drive/MyDrive/PRISM AI/adapters/prism_lora
```

B9. Test — across multiple formats, same adapter

```
FastLanguageModel.for_inference(model)
```

```
def generate(source_text: str, output_format: str) -> str:
    instruction = FORMAT_INSTRUCTIONS[output_format]
    messages = [
        {"role": "system", "content": instruction},
        {"role": "user", "content": source_text},
    ]
    inputs = tokenizer.apply_chat_template(
        messages, tokenize=True, add_generation_prompt=True, return_tensors="pt"
    ).to("cuda")
    outputs = model.generate(input_ids=inputs, max_new_tokens=512, temperature=0.7, do_sample=True)
    return tokenizer.decode(outputs[0][inputs.shape[1]:], skip_special_tokens=True)

test_source = "Paste a real short source document here – something NOT in your training data."

for fmt in ["advisory", "linkedin", "executive_summary"]:
    print(f"\n=== {fmt} ===")
    print(generate(test_source, fmt))
```

The attention mask is not set and cannot be inferred from input because pad token is same as eos token. As a consequence, y
Both `max\_new\_tokens` (=512) and `max\_length` (=32768) seem to have been set. `max\_new\_tokens` will take precedence. Please r

=== advisory ===
Both `max\_new\_tokens` (=512) and `max\_length` (=32768) seem to have been set. `max\_new\_tokens` will take precedence. Please r
Summary: On October 12, 2023, Meridian Logistics detected unauthorized access to its customer database via a compromised thi

Threat Details: The intrusion began on October 10 when attackers exploited weak authentication on a vendor integration point

Impact: Customers in North America and Europe are primarily affected. Meridian has notified all affected individuals and is

Recommendations: Immediately audit all third-party API integrations for MFA enforcement. Patch any legacy systems connected

=== linkedin ===
Both `max\_new\_tokens` (=512) and `max\_length` (=32768) seem to have been set. `max\_new\_tokens` will take precedence. Please r
The 2023 Global Water Security Report released by the International Hydrological Institute highlights a critical shift in re

=== executive_summary ===
Q3 Supply Chain Disruption Report: Meridian Logistics Inc.

Date: October 12, 2023

Subject: Impact of Port congestion at Rotterdam and Antwerp on inbound electronics shipments.

Key Findings:

1. Operational Delay: Due to persistent congestion at the Port of Rotterdam and Antwerp, average inbound transit time for el
2. Financial Implications: The delay resulted in an estimated \$4.5 million in expedited freight costs and a 12% reduction in
3. Mitigation Actions: Meridian activated secondary air freight routes for critical SKUs, incurring an additional \$1.2 milli
4. Future Projections: If congestion persists through November, we project a 15% decline in overall Q4 production capacity a

Recommendations:

1. Accelerate contract negotiations with Hamburg and Antwerp ports to secure dedicated lanes.
2. Increase buffer inventory by 20% for high-demand SKUs before the end of Q4.
3. Review third-party carrier agreements to identify potential cost-sharing mechanisms for future delays.

Prepared by: Sarah Jenkins, Senior Operations Manager